

Highlights

Log-Weighted Cross-Entropy Framework for Imbalanced Classification

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- A logarithmic loss that compresses the rare-to-frequent weight ratio of WCE.
- A single-parameter generalization tuned automatically on a small proxy subset.
- An ϵ -smoothed variant keeps per-class weights bounded even for the rarest classes.
- Evaluation spans image, text, tabular, and network-intrusion domains.
- The proposed losses improve as imbalance grows across four evaluation domains.

Log-Weighted Cross-Entropy Framework for Imbalanced Classification

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ARTICLE INFO

Keywords:

Class imbalance
Long-tailed learning
Cross-entropy loss
Cost-sensitive learning
Weight smoothing
Network intrusion detection

ABSTRACT

Class imbalance is a pervasive obstacle in machine learning, arising whenever a few majority categories dominate the training data while critically important minority categories remain severely underrepresented. Among loss-level remedies, Weighted Cross-Entropy (WCE) is the most direct: it scales the loss of each class by the inverse of its frequency. However, as the imbalance becomes more severe, these inverse-frequency weights induce an increasingly large rare-to-frequent weight ratio, so that minority samples can dominate the class-dependent component of the gradient scale and destabilize optimization. More elaborate alternatives such as Class-Balanced (CB) Loss and Focal Loss introduce additional hyperparameters and can be sensitive to their selection and to dataset characteristics. We propose Log-Weighted Cross-Entropy (LWCE), which replaces the inverse-frequency weight with a logarithmically compressed counterpart that keeps a meaningful boost for rare classes while bounding weight growth, and which introduces no additional hyperparameters. We further generalize it to Power-Log Weighted Cross-Entropy (PLWCE) through a single exponent that adjusts the focusing strength and is selected automatically on a small proxy subset, and we introduce a complementary ϵ -smoothed variant (ES-LWCE) that adds a small constant to the logarithmic denominator, providing an explicit, user-controlled cap on the per-class weight and keeping it well defined even for vanishing class counts, such as classes absent from a mini-batch. To assess generality, we evaluate the proposed losses across four distinct application domains that together span a wide range of imbalance severity and model architectures: long-tailed image classification, tabular classification benchmarks, network intrusion detection, and text classification with a Transformer encoder. Across these domains, LWCE and PLWCE improve over Cross-Entropy as the imbalance grows and, unlike the competing re-weighting losses, avoid the severe degradation below plain Cross-Entropy that these baselines occasionally exhibit. These results indicate that logarithmic weight compression offers a simple, stable, and parameter-efficient alternative to existing re-weighting strategies for class-imbalanced classification.

1. Introduction

Across a wide range of real-world classification tasks—including medical diagnosis, financial fraud detection, network intrusion detection, and visual recognition—the categories of interest are rarely observed in equal proportions. Instead, a small number of majority categories concentrate the overwhelming majority of the available samples, while many minority categories are represented by only a handful of examples (He and Garcia, 2009; Johnson and Khoshgoftaar, 2019). This long-tailed structure is not an artifact of poor data collection but an intrinsic property of the underlying phenomena: healthy patients vastly outnumber those with a rare disease, legitimate transactions dwarf fraudulent ones, and benign network traffic far exceeds malicious traffic. Because the minority categories are often precisely the ones that carry the greatest practical cost when missed, building classifiers that remain reliable under imbalance is a problem of broad and lasting importance.

The central difficulty is that the standard training objective is itself biased toward the majority. When a model is trained with ordinary Cross-Entropy (CE) loss, the aggregate gradient signal is dominated by the numerically abundant classes, so the optimizer can drive the overall loss down while largely ignoring rare categories. The resulting classifier attains a high overall accuracy yet performs poorly on exactly the minority classes that matter most—a failure mode

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commonly described as majority bias. Standard accuracy therefore masks the problem rather than revealing it, and a method that addresses imbalance must improve minority-class performance without sacrificing the majority.

Existing remedies fall broadly into two families. Resampling methods reshape the training distribution directly: oversampling techniques such as the Synthetic Minority Over-sampling Technique (SMOTE) synthesize new minority instances (Chawla, Bowyer, Hall and Kegelmeyer, 2002), while undersampling discards majority examples to restore balance. Both directions carry well-known risks—oversampling can introduce synthetic artifacts and encourage overfitting, whereas undersampling discards potentially informative majority data (Drummond and Holte, 2003). Loss-level methods instead modify the training objective. These include margin-based losses that enlarge the decision boundary around minority classes (Cao, Wei, Gaidon, Aréchiga and Ma, 2019), logit-adjustment techniques that calibrate class priors, and loss re-weighting methods that scale the per-class loss by class-dependent weights. Among these, re-weighting is especially attractive for three practical reasons. Unlike resampling, it leaves the training data untouched and therefore introduces neither synthetic artifacts nor information loss; unlike architecture-level remedies, it adds no parameters and can be dropped into virtually any network without altering the model; and it requires only a minimal, modality-independent change to the loss function. Indeed, although many sophisticated long-tailed learning methods have since been proposed, re-weighting remains one of the few approaches that can be applied to almost any architecture without modifying the model itself. For these reasons this paper focuses on the re-weighting family.

Despite its appeal, re-weighting remains unsatisfying in practice. Weighted Cross-Entropy (WCE), the most direct re-weighting scheme, assigns each class a weight inversely proportional to its frequency,

$$w_c^{\text{WCE}} \propto \frac{1}{n_c}. \quad (1)$$

As the minority count decreases relative to the majority count, inverse-frequency weighting induces an increasingly large rare-to-frequent weight ratio, which can cause minority samples to dominate the class-dependent component of the gradient scale and destabilize optimization under extreme imbalance. More sophisticated alternatives were designed to soften this behavior but bring their own drawbacks. Focal Loss reduces the influence of easy, typically majority, examples through a tunable focusing term, but it operates on per-sample confidence rather than class frequency, and its behavior can be sensitive to the focusing parameter and the imbalance level. Class-Balanced (CB) Loss re-weights by an effective number of samples governed by an additional hyperparameter whose effect can be sensitive to the dataset. These methods are effective in many settings, but they alleviate the instability through additional mechanisms rather than by directly controlling the rare-to-frequent weight ratio while preserving the simplicity of WCE. In short, the field still lacks a re-weighting loss that is simultaneously simple, hyperparameter-light, numerically stable, and dependable across both the level of imbalance and the application domain.

To address this gap, we propose Log-Weighted Cross-Entropy (LWCE). The key idea is to replace the inverse-frequency weight of WCE with a logarithmically compressed counterpart, applying a logarithm to the class count before inverting it. This preserves a meaningful emphasis on rare classes—the weight still decreases with class size—but fundamentally bounds how fast that emphasis can grow, so the explosive behavior of WCE is removed. Crucially, the resulting loss introduces no additional hyperparameters and can be used as a drop-in replacement for Cross-Entropy. We further develop LWCE in two complementary directions. The first, Power-Log Weighted Cross-Entropy (PLWCE), controls *how strongly* rare classes are emphasized: it introduces a single exponent that continuously adjusts the focusing strength, ranging from near-uniform weighting to an aggressive emphasis on the tail. This exponent is the only free parameter, and it is selected automatically through a lightweight grid search on a small proxy subset, adding negligible computational overhead. The second, ϵ -Smoothed Log-Weighted Cross-Entropy (ES-LWCE), instead adds a small constant ϵ to the logarithmic denominator and reduces to LWCE as $\epsilon \rightarrow 0$. The base LWCE weight is already bounded for every class that appears at least once, so the role of ES-LWCE is not to make an otherwise unbounded weight finite, but to provide an *explicit, user-controlled* cap of $1/\epsilon$ on the per-class weight and to keep it well defined in the degenerate case where a class is temporarily absent—for example, when class frequencies are estimated per mini-batch or maintained online for streaming or growing label sets—without changing the relative ordering of class weights. It is therefore best viewed as a robustness extension for such vanishing- or drifting-count settings rather than as a separate competitor on datasets with fixed, strictly positive counts. The two extensions are otherwise orthogonal: PLWCE adjusts the *strength* of the re-weighting, whereas ES-LWCE controls its *maximum weight* and smooths the extreme tail. Although the two can in principle be combined into a single power-log-smoothed weight, we study them separately to isolate the effect of focusing strength from that of smoothing.

We assess the proposed losses through a deliberately broad empirical study rather than a single benchmark. To test whether the benefit of logarithmic compression generalizes, we evaluate across three application domains that together cover a wide spectrum of imbalance severity: long-tailed image classification on standard vision benchmarks, a collection of tabular classification datasets drawn from public repositories, and several network intrusion detection datasets in which rare attack types are extremely scarce. In every domain we compare the same set of re-weighting losses under a controlled protocol with repeated runs, and we report metrics that explicitly reflect minority-class performance rather than overall accuracy alone. This cross-domain design lets us examine not only whether the proposed losses help, but how their advantage changes as imbalance becomes more extreme.

The main contributions of this paper are as follows.

1. We propose Log-Weighted Cross-Entropy (LWCE), a parameter-free drop-in replacement for Weighted Cross-Entropy that substitutes the inverse-frequency weight with a logarithmically compressed one, reducing the dynamic range of the class weights and the resulting optimization instability while retaining a meaningful emphasis on rare classes.
2. We develop two complementary extensions of LWCE: Power-Log Weighted Cross-Entropy (PLWCE), whose single exponent—tuned automatically on a small proxy subset—controls the focusing strength, and ϵ -Smoothed Log-Weighted Cross-Entropy (ES-LWCE), whose smoothing constant imposes an explicit, user-controlled cap on the per-class weight and keeps it well defined even for vanishing or mini-batch-level class counts.
3. We analyze the theoretical properties of the proposed weighting functions, showing that the weights are bounded for positive class counts, that ES-LWCE remains well defined even at vanishing counts, and that their rare-to-frequent weight ratios grow logarithmically (LWCE) or polylogarithmically (PLWCE) rather than linearly with the imbalance ratio.
4. We provide a controlled cross-domain evaluation spanning image, tabular, network-intrusion, and text classification data, showing that the proposed losses improve over Cross-Entropy as imbalance grows and, unlike Class-Balanced Loss and Focal Loss, avoid the severe degradation below plain Cross-Entropy that these baselines occasionally exhibit.

The remainder of this paper is organized as follows. Section 2 reviews related work on learning from imbalanced and long-tailed data. Section 3 defines the proposed losses and analyzes their properties. Section 4 describes the experimental setup across the three domains and presents the results. Section 5 concludes and discusses limitations and future work.

2. Related Work

Methods for learning under class imbalance can be organized by the level at which the imbalance is addressed: the *data*, by reshaping the training distribution; the *loss*, by making the training objective cost-sensitive; the *representation*, by rebalancing the learned feature space or classifier; and the *architecture*, by restructuring the model itself (He and Garcia, 2009; Johnson and Khoshgoftaar, 2019; Krawczyk, 2016; Branco, Torgo and Ribeiro, 2016; Zhang, Kang, Hooi, Yan and Feng, 2023; Chen, Yang, Yu, Shi and Chen, 2024). Table 1 summarizes this taxonomy with representative methods. Our proposal is a loss-level method and, more specifically, a frequency-based re-weighting loss. We review all four levels for context and then position logarithmic re-weighting among existing cost-sensitive losses.

Table 1

A taxonomy of approaches to class-imbalanced learning, organized by the level at which the imbalance is addressed. The proposed losses are loss-level, frequency-based re-weighting methods.

Level	Representative methods
Data-level	SMOTE, ADASYN, under-/over-sampling
Loss-level	WCE, CB Loss, Focal Loss, LWCE (ours)
Representation-level	Decoupled training, distribution alignment
Architecture-level	Multi-branch, multi-expert models

2.1. Data-level resampling

Resampling rebalances the class distribution before or during training. Oversampling synthesizes additional minority instances—the Synthetic Minority Over-sampling Technique (SMOTE) interpolates between neighbouring minority samples (Chawla et al., 2002), and ADASYN adapts the amount of synthesis to local difficulty (He, Bai, Garcia and Li, 2008)—whereas undersampling discards majority examples and can outperform oversampling on classical learners (Drummond and Holte, 2003). Although effective and classifier-agnostic, synthetic oversampling may introduce noisy or unrealistic samples and encourage overfitting, while undersampling discards potentially informative data (Buda, Maki and Mazurowski, 2018; Branco et al., 2016). These trade-offs motivate objective-level alternatives that leave the data untouched.

2.2. Cost-sensitive and re-weighting losses

Algorithm-level methods instead make the loss itself imbalance-aware. We group them by the signal that drives the re-weighting into frequency-based, difficulty-based, and margin- or logit-based methods. *Frequency-based* methods scale each class loss by a function of its count. The most direct instance is Weighted Cross-Entropy, which sets $w_c \propto 1/n_c$ so that the weight is inversely proportional to class count (Wang, Liu, Wu, Cao, Meng and Chen, 2019); these weights, however, produce a rare-to-frequent ratio that grows linearly with the imbalance ratio and can destabilize optimization, and the benefit of such importance weighting is known to diminish for over-parameterized networks driven to low training error (Byrd and Lipton, 2019). Class-Balanced Loss softens this by weighting with an effective number of samples, $w_c \propto (1 - \beta)/(1 - \beta^{n_c})$, but this introduces an extra hyperparameter $\beta \in [0, 1)$ (Cui, Jia, Lin, Song and Belongie, 2019). *Difficulty-based* methods modulate each example by its own hardness rather than by its class: Focal Loss down-weights easy, mostly majority, samples by a factor $(1 - p_t)^\gamma$, where p_t is the predicted probability of the true class and γ a tunable focusing exponent (Lin, Goyal, Girshick, He and Dollár, 2017), and the Gradient Harmonizing Mechanism flattens the distribution of gradient norms so that both easy samples and hard outliers are suppressed (Li, Liu and Wang, 2019). *Margin- and logit-based* methods reshape the decision geometry rather than the per-class loss magnitude: they enlarge minority margins (Cao et al., 2019), equalize gradients across frequencies (Tan, Wang, Li, Li, Ouyang, Yan and Yan, 2020), adjust logits by class priors (Menon, Jayasumana, Rawat, Jain, Veit and Kumar, 2021), or learn the weights themselves through meta-learning and cost-sensitive representation learning (Ren, Zeng, Yang and Urtasun, 2018; Ren, Yu, Sheng, Ma, Zhao, Yi and Li, 2020; Khan, Hayat, Bennamoun, Sohel and Togneri, 2018), with recent work casting weight learning as an optimal-transport problem between the imbalanced and a balanced distribution (Guo, Li, Zheng, Zhao, Zhou and Zha, 2022). These losses improve over plain Cross-Entropy in many settings, but they introduce additional, often sensitive, hyperparameters and can behave inconsistently across datasets. Our logarithmic family belongs to the frequency-based group and is therefore complementary to the difficulty- and margin-based axes; it targets the central weakness of frequency-based re-weighting by bounding the frequency weight without introducing sensitive hyperparameters.

2.3. Representation learning and decoupling

A complementary line of work argues that imbalance harms the classifier more than the feature extractor. Decoupling representation and classifier learning (Kang, Xie, Rohrbach, Yan, Gordo, Feng and Kalantidis, 2020) and distribution alignment (Zhang, Li, Yan, He and Sun, 2021) rebalance only at the classification stage, while open long-tailed recognition addresses imbalance jointly with open-set and few-shot effects (Liu, Miao, Zhan, Wang, Gong and Yu, 2019); recent surveys provide a broad taxonomy of these directions (Zhang et al., 2023). This concern extends to the era of large pretrained and foundation models, where the fine-tuning strategy itself interacts with the long tail and must be adapted to avoid harming the rarest classes (Shi, Wei, Zhou, Shao, Han and Li, 2024). Such methods are largely orthogonal to the choice of loss and can be combined with re-weighting, so our contribution—confined to the loss term—composes with them rather than competing.

2.4. Architecture-level and multi-expert methods

A third family addresses imbalance by restructuring the network rather than the data or the loss. Multi-branch designs train separate pathways for the head and tail of the distribution and fuse them through cumulative, bilateral learning (Zhou, Cui, Wei and Chen, 2020). Mixture-of-experts and multi-expert approaches extend this idea by training several diverse, distribution-aware experts and then routing or aggregating their predictions: RIDE routes inputs to complementary experts (Wang, Lian, Miao, Liu and Yu, 2021), ACE assembles ally-complementary experts in a single stage (Cai, Wang and Hwang, 2021), LFME distils knowledge from multiple experts trained on balanced

subsets (Xiang, Ding and Han, 2020), and test-agnostic aggregation adapts the expert ensemble to an unknown test distribution (Zhang, Hooi, Hong and Feng, 2022). Such designs can deliver strong tail accuracy, but they increase parameter count and inference cost, and—like the resampling and representation methods above—they are orthogonal to the training objective: a re-weighting loss such as ours can be applied within each branch or expert rather than replacing the architecture.

2.5. Positioning of the proposed losses

Within the re-weighting family, existing schemes face a dilemma: inverse-frequency weighting is parameter-free but its rare-to-frequent weight ratio grows without control, whereas more stable variants regain control only by adding sensitive hyperparameters (Lin et al., 2017; Cui et al., 2019). Logarithmic compression offers a middle ground that, to our knowledge, has not been systematically studied as a re-weighting principle across application domains. LWCE realizes it without any additional hyperparameter, PLWCE adds a single exponent that tunes the focusing strength, and ES-LWCE adds an ϵ term that keeps the weight bounded and well defined for the rarest classes. Table 2 contrasts these properties with the main cost-sensitive baselines: WCE is parameter-free but its rare-to-frequent weight ratio is uncontrolled, Focal and CB Loss regain stability only at the cost of an extra hyperparameter, whereas LWCE is a frequency-based, parameter-free scheme whose weight ratio stays within a controlled range. We evaluate this family against the representative cost-sensitive losses above under a unified protocol spanning image, tabular, network-intrusion, and text classification domains—each of which sustains its own active line of imbalance research, from dedicated tabular benchmarks (Liu, Li, Yang, Wei, Kang, Zhu, Hamann, He and Tong, 2025) to intrusion detection under severe class skew (Abdelkhalek and Mashaly, 2023).

Table 2

Positioning of LWCE against representative cost-sensitive losses. “Signal” is the quantity that drives the re-weighting; “Extra parameter” indicates whether the loss needs a tuned hyperparameter; “Frequency weight” states how the class frequency enters the loss (a multiplicative class weight, an additive logit shift, or not at all); and “Controlled range” indicates whether the class-frequency weight is kept within a controlled range under extreme imbalance. For Focal Loss the only bounded factor is the per-sample difficulty modulation $(1 - p_i)^{\gamma} \in [0, 1]$, not a class-frequency weight; Logit Adjustment applies the class prior additively to the logits, so the criterion does not apply (n/a). LWCE compresses the class-frequency weight itself.

Method	Signal	Extra parameter	Frequency weight	Controlled range
WCE	Frequency	No	Multiplicative	No
Focal Loss	Difficulty	Yes	No	Difficulty factor only
CB Loss	Frequency	Yes	Multiplicative	Yes
Logit Adjustment	Prior	Yes	Additive	n/a
LWCE (ours)	Frequency	No	Multiplicative	Yes

3. Proposed Method

We study a single family of re-weighting losses built on logarithmic weight compression. All three members replace the inverse-frequency weight of WCE with a logarithmically compressed counterpart and differ only in how that base weight is subsequently shaped. The three formulations and the distinct role of each are summarized in Table 3 and defined precisely below.

3.1. The logarithmic re-weighting family

Consider a classification problem with C classes, where class c contains n_c training samples. All three losses re-weight the standard cross-entropy objective by a class-dependent factor w_c ,

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N w_{y_i} \log p_{i,y_i}, \quad (2)$$

where p_{i,y_i} is the predicted probability of the true class y_i of sample i and N is the number of samples; the members differ only in the choice of w_c . Log-Weighted Cross-Entropy (LWCE) uses the parameter-free weight

$$w_c^{\text{LWCE}} = \frac{1}{\log(1 + n_c)}. \quad (3)$$

Power-Log Weighted Cross-Entropy (PLWCE) raises the logarithmic denominator to a single exponent $\alpha > 0$,

$$w_c^{\text{PLWCE}} = \frac{1}{(\log(1 + n_c))^\alpha}, \quad (4)$$

which continuously tunes how strongly rare classes are emphasized and reduces to LWCE at $\alpha = 1$. ε -Smoothed Log-Weighted Cross-Entropy (ES-LWCE) adds a small constant $\varepsilon > 0$ to the denominator,

$$w_c^{\text{ES-LWCE}} = \frac{1}{\log(1 + n_c) + \varepsilon}, \quad (5)$$

which caps the weight at $1/\varepsilon$.

Weight normalization. In all three formulations the raw weights are mean-normalized before use,

$$\tilde{w}_c = \frac{C w_c}{\sum_{j=1}^C w_j}, \quad (6)$$

so that $\frac{1}{C} \sum_c \tilde{w}_c = 1$ and the overall loss scale is independent of the weighting scheme; only the *relative* emphasis across classes differs. We apply the identical normalization to every re-weighting baseline, including WCE and Class-Balanced loss, so that all methods share one scale convention and differences in performance cannot be attributed to a global change in effective learning rate. All properties below are stated for the raw weights w_c ; since Eq. (6) is a single positive rescaling common to every class, it leaves the rare-to-frequent *ratios* of Propositions 3 and 4 unchanged.

Table 3

The proposed logarithmic re-weighting family. n_c denotes the number of training samples in class c ; $\alpha > 0$ and $\varepsilon > 0$ are the only free parameters, belonging to PLWCE and ES-LWCE respectively.

Loss	Class weight w_c	Role
LWCE	$\frac{1}{\log(1 + n_c)}$	Parameter-free base weight
PLWCE	$\frac{1}{(\log(1 + n_c))^\alpha}$	Exponent α tunes emphasis strength
ES-LWCE	$\frac{1}{\log(1 + n_c) + \varepsilon}$	Constant ε bounds and smooths the weight

3.2. Theoretical properties

We now make precise the sense in which logarithmic compression tames the weight explosion of WCE. Throughout, let $\rho = n_{\max}/n_{\min} \geq 1$ denote the imbalance ratio between the most and least frequent classes, and let the subscripts min and max index the rarest and the most frequent class, respectively. Four properties characterize the proposed family.

Proposition 1 (Monotonicity). For LWCE, PLWCE, and ES-LWCE, the weight w_c is strictly decreasing in the class count n_c . Consequently a rarer class always receives a larger weight, which preserves the qualitative behaviour of inverse-frequency weighting. *Proof.* $\log(1 + n_c)$ is strictly increasing in n_c , so its reciprocal, any positive power of its reciprocal, and the reciprocal of its ε -shift are all strictly decreasing in n_c . $\square$ Because mean normalization (Eq. (6)) multiplies every class weight by the same positive constant, it preserves this ordering, so the normalized weights $\tilde{w}_c$ are also strictly decreasing in n_c .

Proposition 2 (Boundedness). For $n_c \geq 1$ the individual raw weights are bounded: $0 < w_c^{\text{LWCE}} \leq 1/\log 2$ and, for any fixed $\alpha > 0$, $0 < w_c^{\text{PLWCE}} \leq 1/(\log 2)^\alpha$. The ES-LWCE weight is bounded even without this assumption: $0 < w_c^{\text{ES-LWCE}} \leq 1/\varepsilon$ for every $n_c \geq 0$. These bounds concern the raw weights w_c ; the normalized weights $\tilde{w}_c$ of Eq. (6) depend on the class composition and are not claimed to satisfy the same fixed bound. *Proof.* Each weight is maximized at its smallest admissible count by Proposition 1. For LWCE and PLWCE this is $n_c = 1$, giving $1/\log 2$ and $1/(\log 2)^\alpha$. For ES-LWCE the denominator is at least ε , attained as $n_c \rightarrow 0$, so the weight never exceeds $1/\varepsilon$. $\square$

This proposition also clarifies the distinct role of ES-LWCE. LWCE is already bounded for every class that appears at least once; what ES-LWCE adds is robustness at $n_c = 0$, where the LWCE denominator $\log(1 + n_c)$ vanishes and the weight diverges. Such vanishing counts arise, for instance, when class frequencies are estimated per mini-batch. ES-LWCE turns the implicit bound of LWCE into an explicit, user-controlled cap of $1/\varepsilon$.

Proposition 3 (Dynamic-range compression). Fix the minority count $n_{\min} \geq 1$ and let $\rho = n_{\max}/n_{\min} \rightarrow \infty$ through growth of $n_{\max}$. The rare-to-frequent weight ratio then grows *linearly* in ρ under WCE but only *logarithmically* for LWCE and *polylogarithmically* for PLWCE. Under WCE,

$$\frac{w_{\min}^{\text{WCE}}}{w_{\max}^{\text{WCE}}} = \frac{n_{\max}}{n_{\min}} = \rho, \quad (7)$$

whereas under LWCE and PLWCE,

$$\frac{w_{\min}^{\text{LWCE}}}{w_{\max}^{\text{LWCE}}} = \frac{\log(1 + n_{\max})}{\log(1 + n_{\min})} = O(\log \rho), \quad \frac{w_{\min}^{\text{PLWCE}}}{w_{\max}^{\text{PLWCE}}} = \left(\frac{\log(1 + n_{\max})}{\log(1 + n_{\min})} \right)^\alpha = O((\log \rho)^\alpha). \quad (8)$$

Here $O(\cdot)$ denotes the asymptotic growth order in the imbalance ratio: for positive functions f and g , $f = O(g)$ means that f grows no faster than a constant multiple of g as $\rho \rightarrow \infty$. Comparing Eqs. (7) and (8), the compression reduces the dynamic range of the weights from linear in ρ (WCE) to *logarithmic* for LWCE and *polylogarithmic*, $O((\log \rho)^\alpha)$, for PLWCE, while preserving the ordering that favours rare classes. Note that this is a statement about the *ratio*: unlike the individual weights of Proposition 2, the ratio is not constant-bounded but grows without limit as $\rho \rightarrow \infty$ —only much more slowly than the linear growth of WCE. For PLWCE with a large exponent α the polylogarithmic ratio can itself become sizeable, so the compression is strongest for LWCE and is moderated as α increases. Figure 1 makes this compression concrete: with the majority count fixed at $n_{\max} = 10^4$, the rarest class receives a weight $\rho = 10^4$ times larger than the majority under WCE, but only about 13 times larger under LWCE, with PLWCE and ES-LWCE interpolating between these regimes.

Proposition 4 (Gradient-scale control). In weighted cross-entropy the per-sample gradient with respect to the logits is the unweighted cross-entropy gradient scaled by the class weight w_{y_i} (Eq. (2)). The full gradient magnitude also depends on the predicted probabilities and the feature representation, so we do not claim that the minority-to-majority gradient ratio equals the weight ratio exactly. Rather, the *class-dependent multiplicative component* of the gradient scale follows the same asymptotic order as the corresponding weight ratio of Proposition 3: it is $O(\rho)$ under WCE but $O(\log \rho)$ under LWCE. Logarithmic compression therefore limits the extent to which minority gradients can overwhelm the majority under extreme imbalance, which is the mechanism by which LWCE mitigates the optimization instability of WCE; ES-LWCE additionally caps the worst-case class-dependent factor at $1/\varepsilon$. Section 4 tests this mechanism directly by tracking the per-epoch gradient norm and the empirical minority-to-majority gradient ratio.

3.3. Parameter selection for PLWCE

LWCE and ES-LWCE require essentially no tuning: LWCE is parameter-free, and the ε of ES-LWCE is a user-set cap rather than an accuracy knob (Section 3.4). PLWCE, however, exposes a genuine hyperparameter α that controls the focusing strength, with three qualitatively distinct regimes:

- $0 < \alpha < 1$: *mild* re-weighting, weaker than LWCE;
- $\alpha = 1$: exactly LWCE;
- $\alpha > 1$: *strong* tail emphasis.

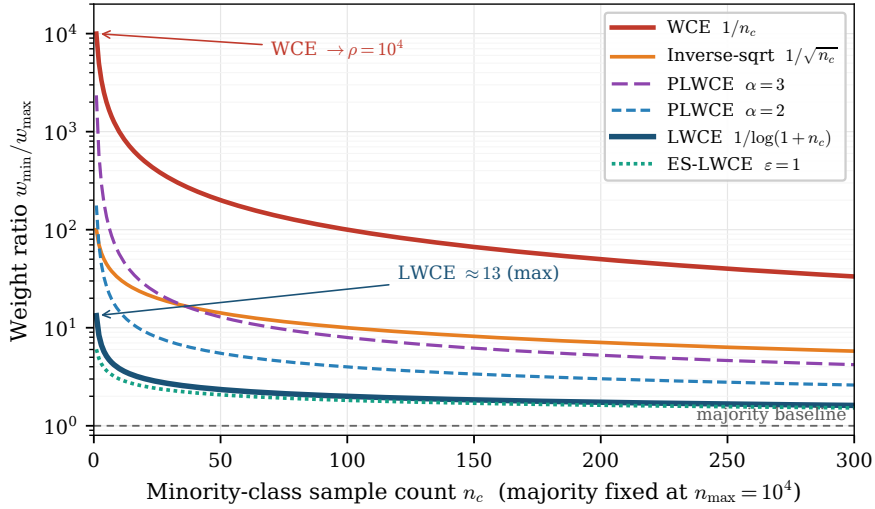


Figure 1: Dynamic-range compression (Proposition 3). The vertical axis is the rare-to-frequent weight ratio $w_{\min}/w_{\max}$ as a function of the minority count n_c , with the majority count fixed at $n_{\max} = 10^4$; this ratio equals the imbalance ratio ρ for WCE and is invariant to the mean-normalization of Eq. (6). Inverse-frequency (WCE) weighting explodes linearly to $\rho = 10^4$ as $n_c \rightarrow 1$, and the inverse-square-root variant to $\sqrt{\rho}$, whereas the logarithmic family stays within a small multiple of the majority baseline (LWCE peaks near 13). ES-LWCE lies at or below LWCE, confirming that its ε term only softens the weight rather than amplifying it.

We select α with a reproducible, data-driven procedure that never touches the test set. For each dataset (and each imbalance level, where applicable) we draw a stratified *proxy* subset from the *training* split—a fixed fraction of the training data, with a minimum number of samples per class so that rare classes are never eliminated from the proxy—and hold out a validation portion of that proxy. A model is trained on the proxy for a reduced number of epochs at each candidate α on a fixed grid, and we retain the value that maximizes the macro-averaged F_1 on the proxy validation set. Only this single scalar is then used to train the final model under the full protocol. The search is therefore confined to the training split, is run once per dataset, and adds a bounded, one-time cost (the number of grid points times the reduced-epoch proxy budget) on top of a single full run; it introduces no per-iteration overhead and no test-set leakage.

3.4. ES-LWCE and practical use cases

ES-LWCE is not intended to outperform LWCE on datasets whose class counts are fixed and strictly positive; there, LWCE is already bounded (Proposition 2) and the two coincide as $\varepsilon \rightarrow 0$. Its value appears when the class counts used to form the weights can *vanish or drift*. When frequencies are estimated per mini-batch, or maintained online for streaming or continually growing label sets, a class may be momentarily absent ($n_c = 0$); the LWCE weight then diverges, whereas ES-LWCE remains well defined with raw weight at most $1/\varepsilon$ *before* the mean normalization of Eq. (6). The constant ε trades off two effects:

- small ε : stronger minority emphasis, closer to LWCE;
- large ε : more conservative gradient scaling, closer to uniform.

As $\varepsilon \rightarrow \infty$ the raw weights all vanish, but after mean normalization they converge to the uniform (Cross-Entropy) weighting; thus, on the normalized weights actually used in training, ε shifts the emphasis monotonically from the LWCE limit toward the uniform limit and never past LWCE, so it acts purely as a smoothing/robustness control. We therefore position ES-LWCE as an extension for vanishing- or batch-adaptive-count settings rather than as a method that is required at the dataset level.

3.5. Computational complexity

All three losses add negligible cost over plain Cross-Entropy. The class weights depend only on the class counts and are precomputed once from the training split; there are no additional model parameters and no extra backward-pass

terms. Once the weights are precomputed, LWCE and ES-LWCE incur the same asymptotic training complexity as WCE, requiring only a scalar multiplication per sample. The only added cost specific to PLWCE is the one-time grid search of Section 3.3, which is bounded and performed on a reduced proxy rather than at full training scale.

4. Experiments

Our experiments are organized around four questions:

1. Does the advantage of the logarithmic family grow as the imbalance becomes more severe?
2. Does LWCE empirically reduce the optimization instability of WCE?
3. Does the PLWCE exponent α control tail emphasis as intended?
4. Is ES-LWCE stable under vanishing or extremely rare class counts?

To probe architecture- and modality-independence, we evaluate the same loss family across image classification (CIFAR-10/100-LT with a ResNet-32 backbone), tabular classification and network-intrusion detection (multilayer perceptrons), and text classification (a compact Transformer encoder on a long-tailed emotion benchmark).

4.1. Datasets and imbalance construction

We evaluate on four domains spanning a wide range of imbalance severity, from mildly skewed tabular benchmarks ($\rho \approx 2$) to extreme network-intrusion data (ρ up to 1.7×10^4); Table 4 lists all datasets. For the two image benchmarks we construct long-tailed (LT) versions of CIFAR-10 and CIFAR-100 by subsampling each class to an exponential profile,

$$n_c = n_{\max} \rho^{-\frac{c-1}{C-1}}, \quad c = 1, \dots, C, \quad (9)$$

with $n_{\max} = 5000$ (CIFAR-10) and 500 (CIFAR-100), and sweep the imbalance ratio $\rho \in \{10, 50, 100\}$; the standard test sets are left balanced. The tabular, network-intrusion, and text datasets are used with their *natural* class imbalance and their official or standard train/test splits. GoEmotions is filtered to single-label examples, giving a 28-class long-tailed text benchmark.

4.2. Models and training protocol

Because the study compares losses rather than architectures, within each domain the backbone and the full training protocol are held identical across all losses; only the loss function changes. *Image*: a ResNet-32 backbone trained from scratch with SGD (learning rate 0.1, momentum 0.9, weight decay 2×10^{-4}), a multi-step schedule decaying by 0.01 at epochs 160 and 180, batch size 128, and standard CIFAR augmentation (random crop with 4-pixel padding and horizontal flip) for 200 epochs. *Tabular* and *network intrusion*: a multilayer perceptron (256–128–64 with batch normalization, ReLU, and dropout 0.3) trained with Adam (learning rate 10^{-3} , weight decay 10^{-4}) and cosine annealing to 10^{-5} , batch size 512, for 100 epochs (tabular) and 50 epochs (network). *Text*: a compact Transformer encoder ($d_{\text{model}} = 128$, 4 heads, 4 layers, feed-forward width 256, masked mean pooling) trained from scratch with AdamW (learning rate 5×10^{-4} , weight decay 10^{-2}), a linear warmup over three epochs followed by cosine decay, gradient-norm clipping at 1.0, and batch size 64, for 50 epochs. Every configuration is run with five random seeds (42–46), and we report the mean and standard deviation. All losses use the mean-normalized weights of Eq. (6), including the WCE and Class-Balanced baselines, so that every method shares one scale convention.

4.3. Baselines and evaluation

We compare the three proposed losses (LWCE, PLWCE, ES-LWCE) against Cross-Entropy, Weighted Cross-Entropy, an inverse-square-root variant, Class-Balanced loss (Cui et al., 2019), Focal loss (Lin et al., 2017), and Logit Adjustment (Menon et al., 2021) as a margin/posterior-shift baseline. Because overall accuracy is uninformative under strong imbalance, we report macro-averaged F_1 and worst-class accuracy as the primary metrics, and additionally balanced accuracy, the geometric mean of per-class recall, minority-class recall, and the full per-class accuracy vector. All results are averaged over five random seeds and reported as mean $\pm$ standard deviation.

Table 4

Datasets, grouped by domain. C is the number of classes and ρ the imbalance ratio (ratio of the largest to smallest class count). For the image benchmarks ρ is imposed by Eq. (9); the other domains use their natural imbalance. Train sizes for the CIFAR-LT rows are the constructed totals (split 80/20 into train/validation) at $\rho = 100$.

Dataset	Domain	C	# Train	ρ
CIFAR-10-LT	Image	10	12,406	10/50/100
CIFAR-100-LT	Image	100	10,847	10/50/100
credit_card_fraud	Tabular	2	199,364	578.5
aps_failure	Tabular	2	60,000	59.0
bank_marketing	Tabular	2	31,647	7.5
telco_churn	Tabular	2	4,930	2.8
german_credit	Tabular	2	700	2.3
secom	Tabular	2	1,096	14.0
credit_card_default	Tabular	2	21,000	3.5
glass	Tabular	6	149	8.8
steel_faults	Tabular	7	1,358	12.4
yeast	Tabular	10	1,038	108.0
page_blocks	Tabular	5	3,831	171.9
NSL-KDD	Network	5	125,973	1,295
UNSW-NB15	Network	10	82,332	841
CICIDS2017	Network	15	210,026	8,750
CICIDS2018	Network	15	166,261	17,500
CIC-DDoS2019	Network	18	208,502	1,957
Bot-IoT	Network	4	140,053	1,366
TON_IoT	Network	10	147,730	47.9
RT-IoT2022	Network	12	86,181	3,313
GoEmotions	Text	28	36,308	328.8

4.4. Hyperparameter selection

A recurring concern with re-weighting losses is that any apparent advantage may stem from tuning rather than from the weighting scheme itself. To rule this out, every baseline that exposes a free parameter is tuned under an identical, automated budget, while LWCE is used exactly as defined in Eq. (3) with *no* search at all. Concretely, the PLWCE and inverse-power exponents, the Focal focusing parameter, the ES-LWCE smoothing constant, and the Logit-Adjustment temperature are each selected on a validation split by the procedure of Section 3.3; Cross-Entropy, WCE, the inverse-square-root variant, and Class-Balanced loss are parameter-free or use their authors' recommended value. Search uses a deterministic grid over each parameter (with the Focal grid lower-bounded at $\gamma = 1$, below which the modulating factor $(1 - p_i)^\gamma$ yields an unstable gradient as $p_i \rightarrow 1$), so that the entire admissible range is covered reproducibly, and the smoothing constant ϵ is searched on a logarithmic grid because its effect on the weights is multiplicative. LWCE therefore competes with no tuning against baselines that each receive a full search.

4.5. Main results

Table 5 summarizes the primary metric, macro-averaged F_1 , aggregated within each domain, together with the mean rank of each loss over all 26 dataset-and-imbalance settings. Two patterns are consistent across domains. First, the logarithmic re-weighting family occupies the top of the overall ranking: LWCE attains the best mean rank (4.08), and the proposed losses together with the inverse-power and inverse-square-root variants form a leading cluster well ahead of plain Cross-Entropy (mean rank 6.69). Second, unbounded inverse-frequency weighting is the weakest method overall: WCE has the worst mean rank (8.19) and Class-Balanced loss the second worst (7.62), consistent with the weight-explosion mechanism of Section 3—as the imbalance grows, the uncontrolled weight ratio of WCE degrades its accuracy below even Cross-Entropy.

No single loss dominates every domain, which is expected given how differently the domains are structured: logit adjustment is strongest on the balanced-test image benchmarks, LWCE on the tabular datasets, PLWCE on the long-tailed text data, and the combined variant—by a small margin—on network intrusion. The logarithmic family

is nonetheless consistently near the top in all four, which is the central claim of a domain-independent re-weighting principle. On worst-class accuracy and the geometric mean of per-class recall the picture is more nuanced. WCE and CB often achieve the highest tail recall (for example, the highest geometric mean on the tabular datasets), but only by sacrificing head accuracy, so their macro- F_1 falls to the bottom of the table; the proposed losses instead improve tail performance while preserving overall accuracy. This trade-off is most visible on the 28-class text benchmark, where PLWCE reaches a geometric mean of 0.240 against 0.073 for Cross-Entropy while also achieving the best macro- F_1 , whereas the higher tail recall of CB comes with the lowest overall accuracy. Full per-dataset tables with all metrics are provided in the supplementary material.

Table 5

Main results. Per-domain macro- F_1 (mean over the datasets and imbalance settings of each domain, five seeds), and the mean rank of each loss over all 26 dataset-and-imbalance settings (lower is better). The three proposed losses are grouped at the bottom. Best in each column in **bold**, second best underlined.

Method	Image	Tabular	Network	Text	Mean rank
CE	0.403	0.714	0.642	0.412	6.69
WCE	0.394	0.683	0.622	0.374	8.19
Inv.-power	0.417	0.725	0.667	0.423	4.19
Inv.-sqrt	0.417	0.723	0.660	0.429	4.65
CB Loss	0.395	0.701	0.629	0.399	7.62
Focal Loss	0.401	0.715	0.663	0.405	6.15
Logit Adj.	0.434	0.720	0.631	0.421	4.31
LWCE	0.412	0.730	0.665	0.419	4.08
PLWCE	<u>0.421</u>	0.719	0.666	0.434	4.73
ES-LWCE	0.411	<u>0.725</u>	<u>0.669</u>	0.419	<u>4.38</u>

4.6. Optimization stability and mechanism

Better macro- F_1 alone does not establish that the logarithmic family stabilizes optimization. We therefore verify the mechanism of Section 3 directly, using the most severe image setting (CIFAR-100-LT at $\rho = 100$, 100 classes) as a representative case and treating optimization stability operationally through three quantities: the dynamic range of the class weights actually used in training, the minority-to-majority gradient ratio, and the smoothness and magnitude of the training-loss and gradient-norm curves.

Weight distribution (Proposition 3). On the mean-normalized weights $\tilde{w}_c$ used in training, the empirical dynamic range matches the theory closely. The largest-to-smallest normalized weight ratio is 100 for WCE and 97 for CB (coefficient of variation ≈ 1.2), 10 for the inverse-square-root variant and 8.4 for the inverse-power variant, but only 3.5 for LWCE, 3.0 for PLWCE, and 2.4 for ES-LWCE (coefficient of variation 0.27–0.39). The logarithmic family thus compresses the weight dynamic range by more than an order of magnitude relative to WCE, exactly as predicted by Proposition 3.

Gradient dynamics. Figure 2 tracks the per-epoch training loss, parameter gradient norm, and minority-to-majority gradient ratio R_G . The gradient ratio (panel c) is where the mechanism is clearest: at the start of training WCE and CB amplify the minority-class gradient by a factor of about 21 relative to the majority, whereas LWCE and PLWCE keep this factor near 2–3 and plain CE near 1. This early amplification is exactly the class-dependent gradient scaling of Proposition 4, and it coincides with a slower, noisier decrease of the training loss (panel a) and a higher gradient norm (panel b, roughly 0.36 for WCE and CB versus 0.16 for LWCE and PLWCE at the end of training) for the unbounded losses. Note that the amplification is a *transient* of the early phase: strong re-weighting eventually overfits the few minority training samples, driving their gradients toward zero, so the WCE ratio falls below one late in training. It is the instability during the critical early epochs—not the converged state—that the logarithmic compression removes, and this is consistent with WCE and CB ranking at the bottom of Table 5 despite their re-weighting intent.

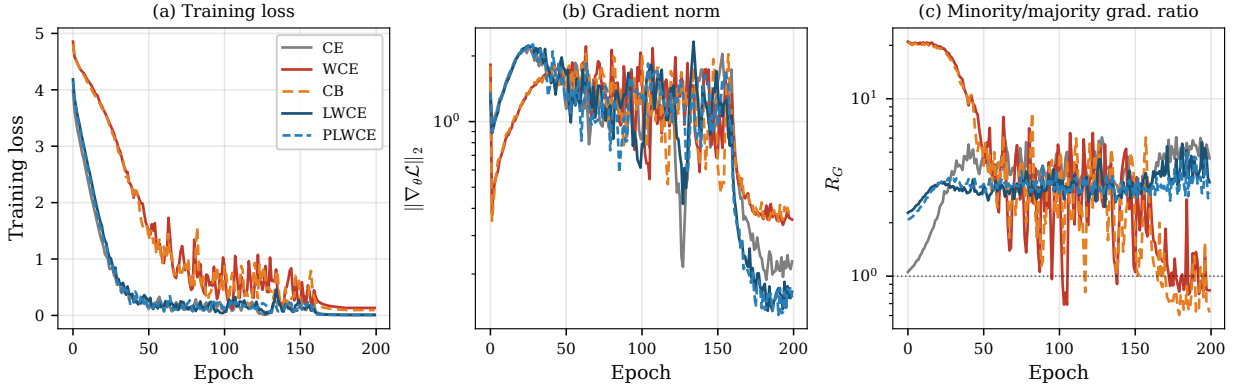


Figure 2: Optimization dynamics on CIFAR-100-LT ($\rho = 100$), averaged over five seeds. (a) Training loss, (b) parameter gradient ℓ_2 -norm $\|\nabla_{\theta} \mathcal{L}\|_2$, and (c) minority-to-majority per-sample logit-gradient ratio R_G (log scale; the dotted line marks $R_G = 1$). Unbounded inverse-frequency losses (WCE, CB) amplify the minority gradient by $\approx 21\times$ early in training and converge more slowly and less smoothly, whereas the logarithmic losses (LWCE, PLWCE) keep the ratio near 2–3 and train stably.

4.7. Ablation and sensitivity

We isolate the contribution of each ingredient of the logarithmic family—the logarithmic compression itself, the power exponent α , and the smoothing constant ϵ —through the ablation of Table 6. Reading from WCE (no compression) to LWCE isolates the effect of logarithmic compression; adding α gives PLWCE and adding ϵ gives ES-LWCE; the *Combined* variant activates both extensions at once, $w_c \propto (\log(1 + n_c) + \epsilon)^{-\alpha}$, and is included—following the same design as the other rows—to probe whether the two extensions are worth merging into a single loss.

The ablation shows that almost all of the benefit comes from the logarithmic compression itself. Averaged over all 26 dataset-and-imbalance settings, replacing the inverse-frequency weight of WCE with the logarithmic weight of LWCE raises macro- F_1 from 0.586 to 0.625, whereas adding the power exponent (PLWCE, 0.623), the smoothing constant (ES-LWCE, 0.624), or both (Combined, 0.624) leaves the aggregate essentially unchanged. The exponent α is not uniformly helpful: it improves the aggregate on the image and text data but not on the tabular datasets, so its effect is dataset-dependent rather than universal, and its value is precisely what the selection procedure of Section 3.3 is designed to recover per dataset.

We do not propose the Combined variant as a method. It yields no improvement over PLWCE or ES-LWCE alone: the selection procedure consistently drives ϵ to the lower end of its grid, so the Combined weight collapses back onto PLWCE. This is a useful negative result rather than a failure—it shows that, on datasets with fixed and strictly positive class counts, the smoothing dimension is inactive for accuracy, exactly as anticipated in Section 3.4: ϵ is a robustness control for vanishing or drifting counts, not an accuracy lever. It therefore supports treating PLWCE and ES-LWCE as separate tools rather than a single combined loss.

Table 6

Ablation of the logarithmic family by ingredient: logarithmic compression, the power exponent α , and the smoothing constant ϵ . The Combined variant activates both extensions and is reported only to test whether merging them helps; it is not proposed as a method. (Metrics averaged over five seeds; full per-domain numbers in the supplementary material.)

Method	Log compression	Power α	Smoothing ϵ	Macro- F_1
WCE	—	—	—	0.586
LWCE	✓	—	—	0.625
PLWCE	✓	✓	—	0.623
ES-LWCE	✓	—	✓	0.624
Combined	✓	✓	✓	0.624

4.8. Statistical analysis

To test whether the ranking of Section 4.5 is statistically meaningful, we treat each of the 26 dataset-and-imbalance settings as a block, average over seeds within each block, and compare the ten losses on macro- F_1 as the primary endpoint. A Friedman test rejects the null hypothesis that the losses are equivalent ($\chi^2 = 60.4$, $df = 9$, $p = 1.1 \times 10^{-9}$). For the post-hoc analysis we take the best-ranked loss, LWCE, as the control and compare it against each other loss with a Wilcoxon signed-rank test over the 26 paired scores, applying a Holm correction for the nine comparisons; Table 7 reports the adjusted p -values together with the mean ranks and the win/tie/loss counts.

LWCE significantly outperforms the four standard baselines—plain Cross-Entropy, WCE, Class-Balanced loss, and Focal loss—at the 5% level after correction (winning, for example, 24 of 26 settings against Cross-Entropy and 20 of 26 against WCE). In particular the method that our theory targets, WCE, is significantly beaten, in line with the mechanism of Section 4.6. The remaining strong re-weighting variants (the inverse-power and inverse-square-root losses, PLWCE, ES-LWCE, and logit adjustment) form a statistical tie with LWCE, with adjusted p -values of 1.0; we therefore do not claim that LWCE is uniformly the single best loss, only that the logarithmic family sits firmly in the top tier and significantly improves on the commonly used baselines while requiring no tuning.

Table 7

Statistical comparison on macro- F_1 over 26 dataset-and-imbalance settings. Mean rank is lower-is-better. The p -value is from a Wilcoxon signed-rank test of LWCE against each loss, Holm-corrected for nine comparisons; win/tie/loss counts are for LWCE versus that loss. Losses that LWCE beats significantly ($p < 0.05$) are marked *.

Method	Mean rank	p (Holm, vs. LWCE)	Win/Tie/Loss
LWCE (control)	4.08	—	—
Inv.-power	4.19	1.000	12/0/14
Logit Adj.	4.31	1.000	11/0/15
ES-LWCE	4.38	1.000	15/0/11
Inv.-sqrt	4.65	1.000	16/0/10
PLWCE	4.73	1.000	17/0/9
Focal Loss	6.15	0.014*	19/0/7
CE	6.69	< 0.001*	24/0/2
CB Loss	7.62	0.010*	20/0/6
WCE	8.19	0.003*	20/0/6

5. Conclusion

CRedit authorship contribution statement

Seung Ho Go: Conceptualization, Methodology, Software, Writing – original draft. **Hyunseok Lee:** Methodology, Validation, Writing – review & editing. **Jin Hee Yoon:** Supervision, Conceptualization, Writing – review & editing.

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